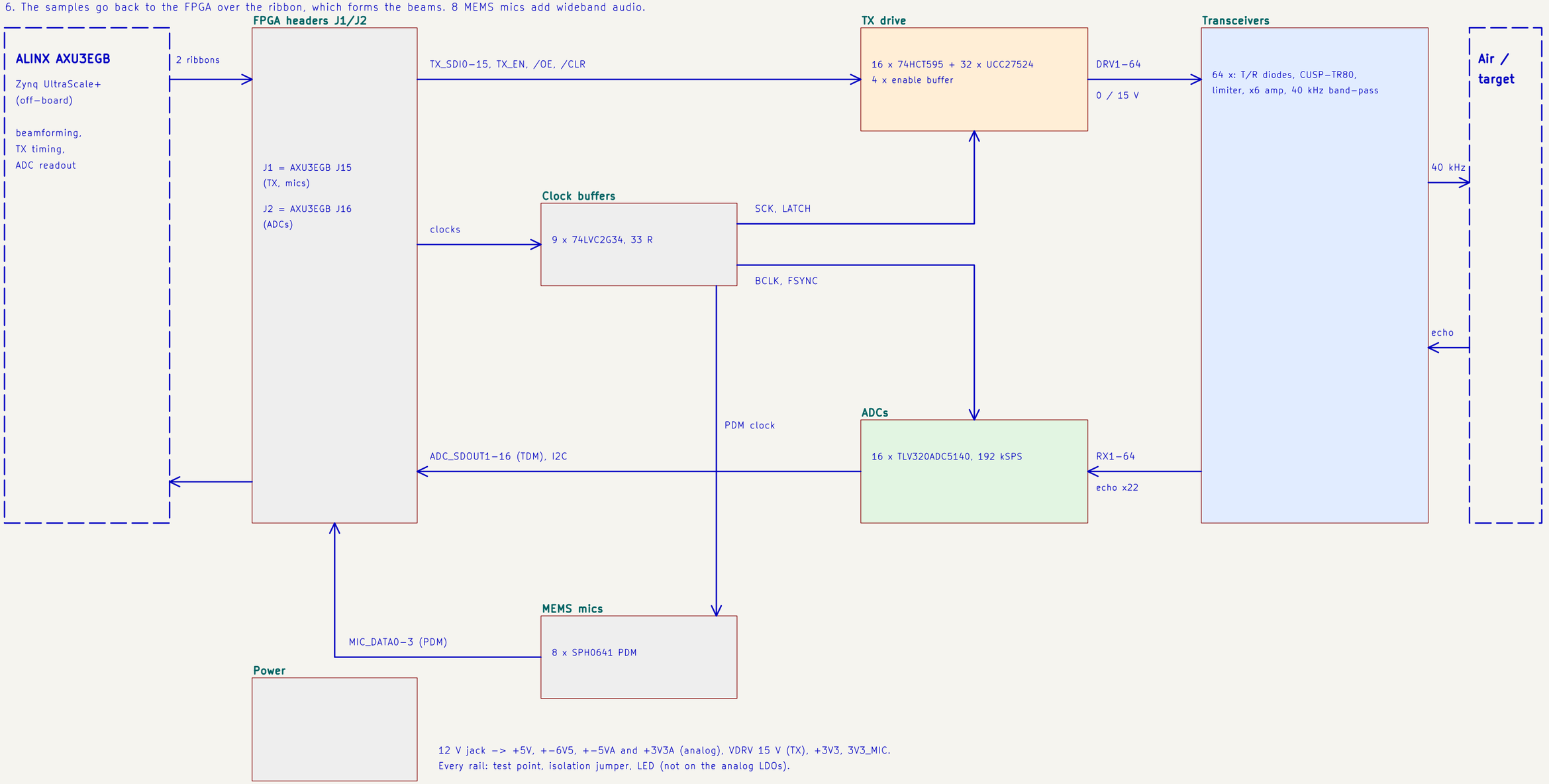
Hybrid 8x8 ultrasonic AESA: how the signal flows

1. The FPGA streams one TX bit per element (16 serial lines, 10 MHz) and the clocks.
2. TX drive: shift registers latch the bits; gate drivers switch each element between 0 and 15 V.
3. Transceivers: the 64 CUSP-TR80 elements emit 40 kHz; echoes return to the same elements.
4. The diode T/R switch and limiter protect the receiver; amplifier + 40 kHz band-pass give x22.
5. ADCs: 16 x TLV320ADC5140 sample all 64 channels at once, 192 kSPS (raw waveform, amplitude and phase).
6. The samples go back to the FPGA over the ribbon, which forms the beams. 8 MEMS mics add wideband audio.



12 V jack -> +5V, +-6V5, +-5VA and +3V3A (analog), VDRV 15 V (TX), +3V3, 3V3_MIC.
Every rail: test point, isolation jumper, LED (not on the analog LDOs).

Click a block to open it. Repeated parts are one sheet each: channel x64, adc x16, tx_group x16, en_buf x4, clk_buf x9, mic x8.

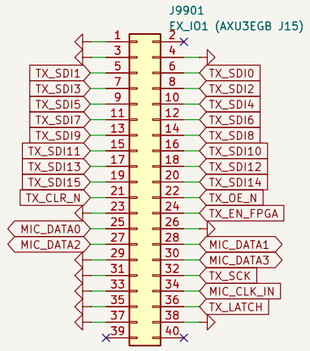
Hybrid 8x8 ultrasonic AESA		
Sheet: /		
File: aesa.kicad_sch		
Title: Hybrid 8x8 ultrasonic AESA: block diagram		
Size: A3	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 1/14

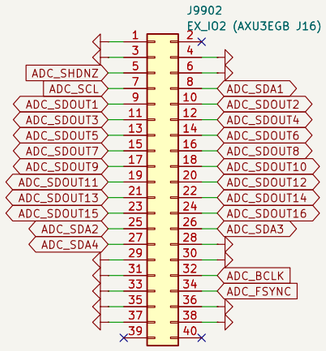
<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

AXU3EGB expansion headers (40-pin IDC, 2.54 mm). J1 = board J15 (TX, mics), J2 = board J16 (ADCs: 16 TDM lines, BCLK, FSYNC, I2C).

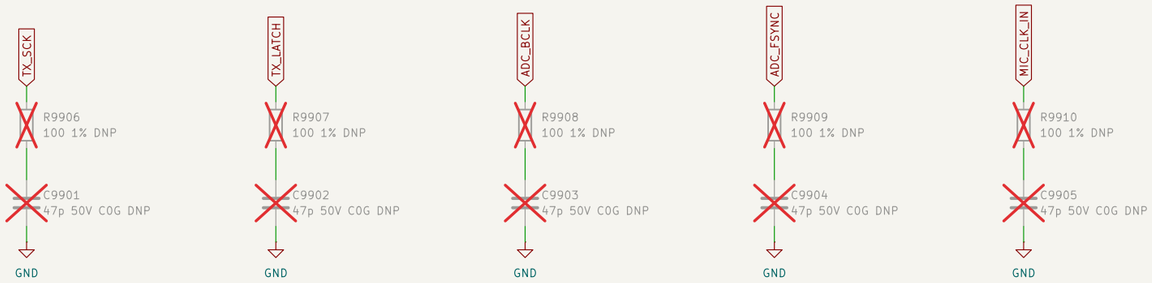
Pin 2 (+5V) and 39/40 (+3.3V) of the FPGA board are left open. Clocks sit between GND conductors (pins 31–37).

Spare FPGA pins tied to GND must stay unused (inputs) in the XDC.

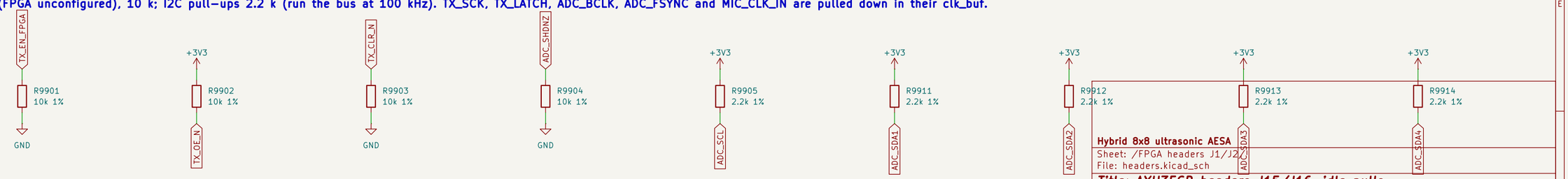




FPGA ribbon outputs: DRIVE 4, SLEW SLOW (near source-matched to the 100-ohm ribbon). DNP fallback: RC end termination 100 R + 47 pF per clock, fit only with DRIVE 8.



Idle pulls (FPGA unconfigured), 10 k; I2C pull-ups 2.2 k (run the bus at 100 kHz). TX_SCK, TX_LATCH, ADC_BCLK, ADC_FSYNC and MIC_CLK_IN are pulled down in their clk_buf.



Hybrid 8x8 ultrasonic AESA

Sheet: /FPGA headers J1/J2/

File: headers.kicad_sch

Title: AXU3EGB headers J15/J16, idle pulls

Size: A3	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 2/14

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

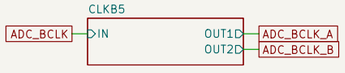
Clock buffers. Each FPGA clock enters one or two clk_buf cells; every output drives at most 4 loads through 33 R.

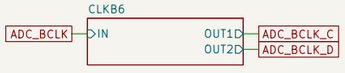


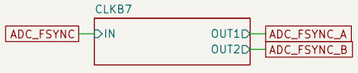


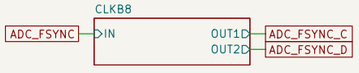














Hybrid 8x8 ultrasonic AESA

Sheet: /Clock buffers/
File: clocks.kicad_sch

Title: Clock buffers: 9 x 74LVC2G34 at the FPGA headers

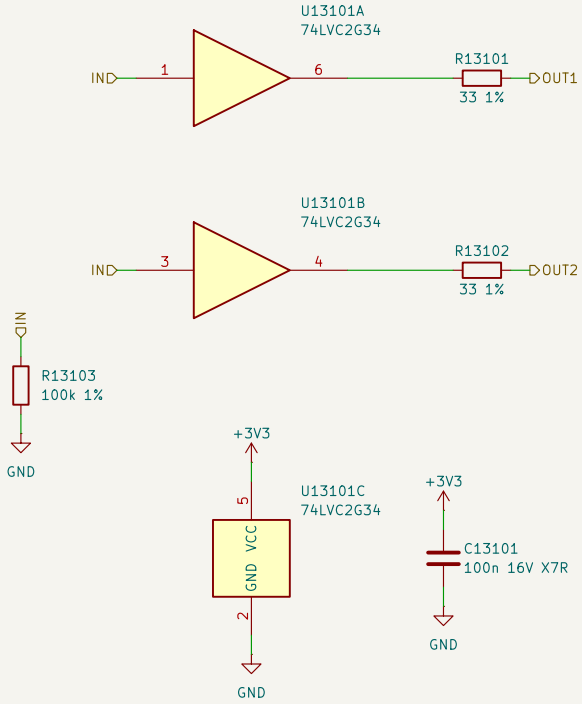
Size: A3	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 3/14

<< up to: Clock buffers: 9 x 74LVC2G34 at the FPGA headers This sheet is used 9 times; the page shows the first instance.

Clock fan-out: 74LVC2G34 at the header, both gates on IN, 33 R series termination per output

Each output drives at most 4 loads (series termination works for a lumped load at the far end).

100 k pull-down keeps the clock low while the FPGA is unconfigured.



Hybrid 8x8 ultrasonic AESA

Sheet: /Clock buffers/CLKB1/

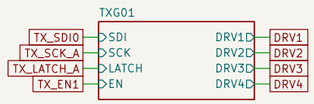
File: clk_buf.kicad_sch

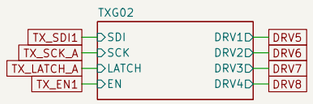
Title: Clock buffer (x9): 74LVC2G34, 33 R outputs

Size: A4	Date: 2026-10-20	Rev: 2
KiCad E.D.A. 10.0.6		Id: 4/14

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

TX drive. The FPGA streams 16 serial lines (TX_SDI0–15) into 16 tx_group cells; each drives 4 elements (DRV). TX_EN_FPGA reaches the drivers through 4 en_buf cells.











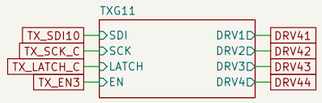


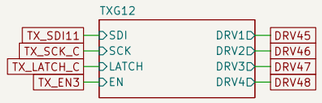


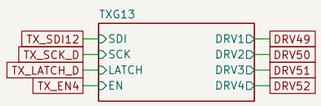


























Hybrid 8x8 ultrasonic AESA

Sheet: /TX drive/
File: tx.kicad_sch

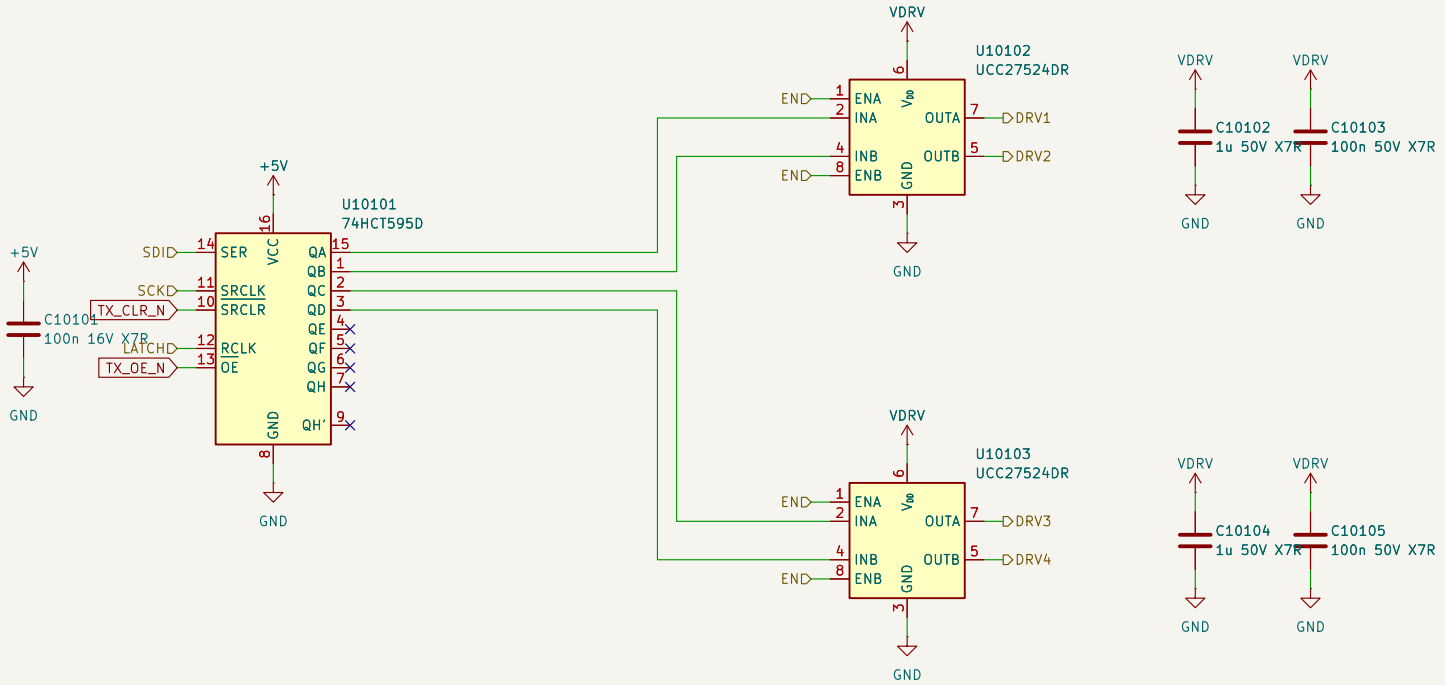
Title: TX drive: 16 shift–register/driver groups, 4 enable buffers

Size: A3	Date: 2026–10–20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 5/14

<< up to: TX drive: 16 shift-register/driver groups, 4 enable buffers

TX group: 4 elements. 74HCT595 QA–QD -> UCC27524 inputs; QE–QH unused (16 data lines, 10 MHz SRCLK)

EN from one 74AHCT1G125 per 8 drivers (en_buf). SCK/LATCH from a clk_buf output (<= 4 loads).

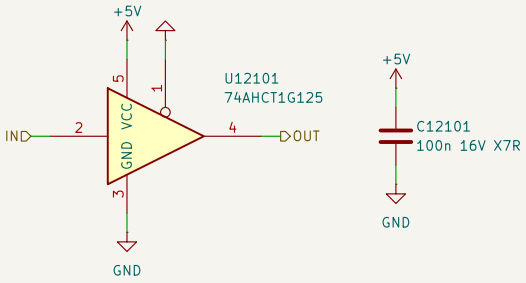


Hybrid 8x8 ultrasonic AESA		
Sheet: /TX drive/TXG01/		
File: tx_group.kicad_sch		
Title: TX group (x16): 74HCT595 + 2 x UCC27524, 4 elements		
Size: A4	Date: 2026-10-20	Rev: 2
KiCad E.D.A. 10.0.6		Id: 6/14

<< up to: TX drive: 16 shift-register/driver groups, 4 enable buffers

This sheet is used 4 times; the page shows the first instance.

TX enable buffer: 74AHCT1G125, 3.3 V in (TTL threshold), 5 V out to 8 UCC27524 (16 EN pins)



Hybrid 8x8 ultrasonic AESA

Sheet: /TX drive/TXEN1/

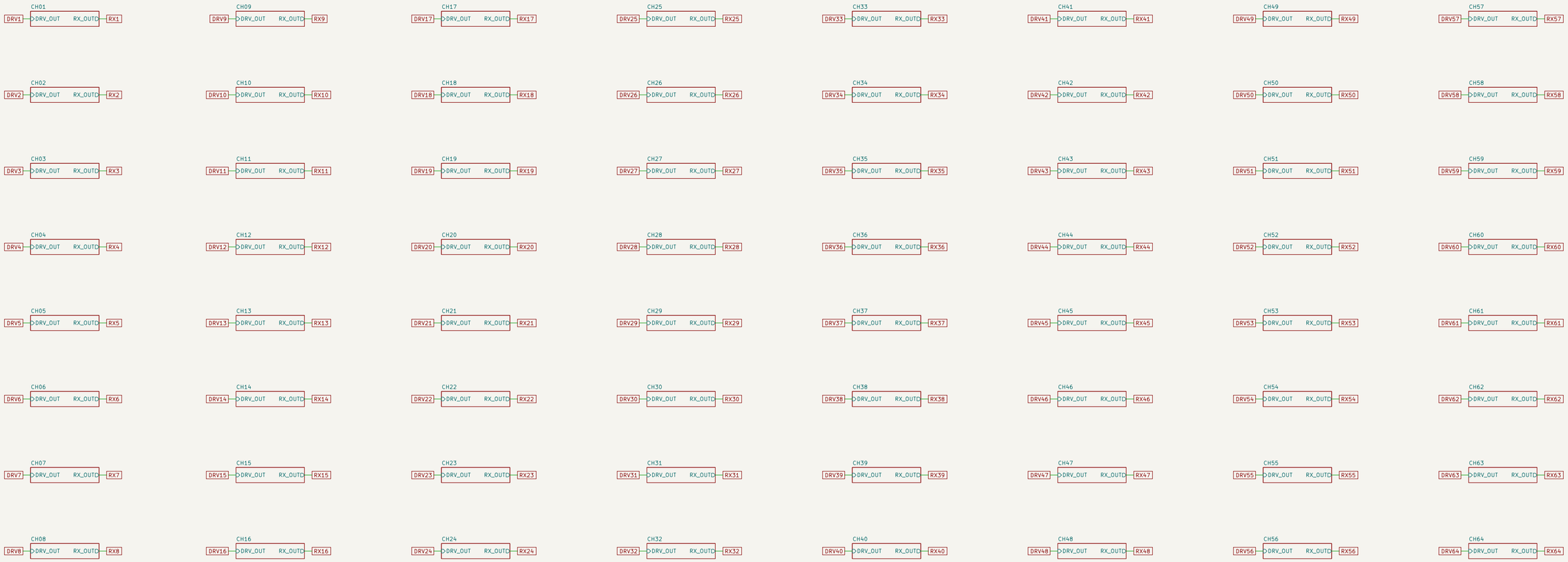
File: en_buf.kicad_sch

Title: TX enable buffer (x4): 74AHCT1G125

Size: A4	Date: 2026-10-20	Rev: 2
KiCad E.D.A. 10.0.6		Id: 7/14

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

Transceiver array. Channel n = element n of the 8 x 8 grid, numbered down the columns (as on the board). Each channel takes DRVn from the TX drive and returns its amplified echo RXn to the ADCs.



Hybrid 8x8 ultrasonic AESA

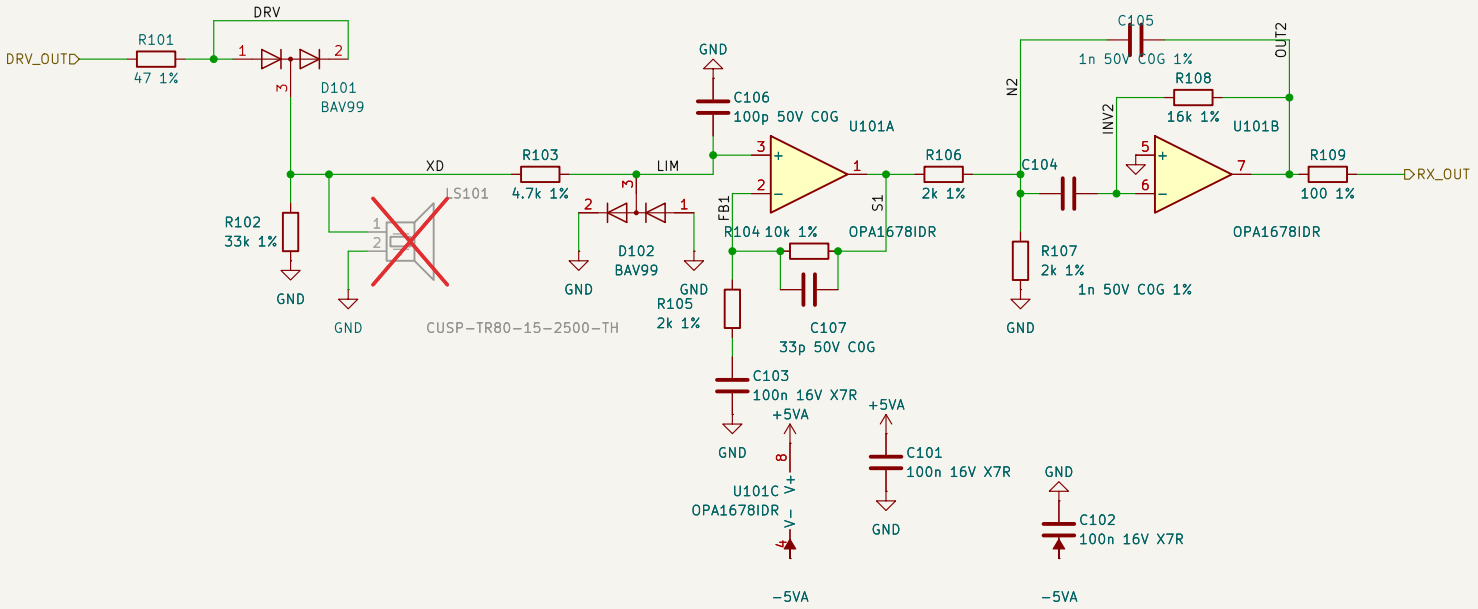
Sheet: /Transceivers/
File: array.kicad_sch

Title: Transceivers: 64 channels (8 x 8 grid)

Size: A2	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 8/14

This sheet is used 64 times; the page shows the first instance.

Transceiver channel (one of 64). TX: UCC27524 half -> 47 R -> BAV99 crossed diodes -> element (0–15 V square).
33k bleed resets the element after TX. RX: 4.7k + BAV99 clamp (+-0.7 V) + 100 pF (340 kHz RF filter).
OPA1678 stage 1: gain 6, AC-coupled at 800 Hz. Stage 2: MFB band-pass f0 39.8 kHz, Q 2, gain -4. Then 100 R to the ADC.
Element -> ADC +27 dB, pass band 31–51 kHz (sim/reports/rx_chain.md). Band-pass parts 1 % C0G for phase match.



Hybrid 8x8 ultrasonic AESA		
Sheet: /Transceivers/CH01/		
File: channel.kicad_sch		
Title: Transceiver channel (x64): TX, T/R switch, RX amplifier		
Size: A4	Date: 2026-10-20	Rev: 2
KiCad E.D.A. 10.0.6		Id: 9/14

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

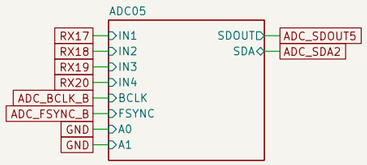
ADCs. ADC k digitises RX 4(k−1)+1 … 4k and returns one TDM line (SDOUTk). Group g (ADC 4g−3 … 4g) shares SDAg and BCLK/FSYNC output g; A1/A0 give each ADC of a group its own I2C address.

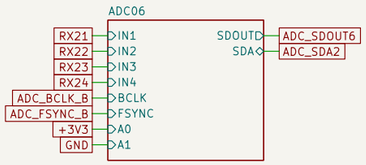


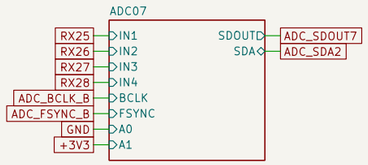












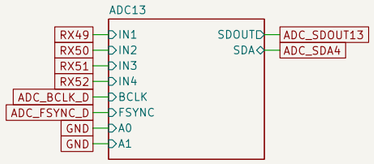




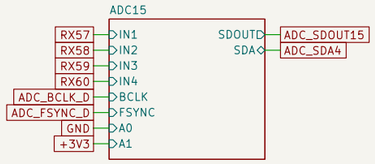


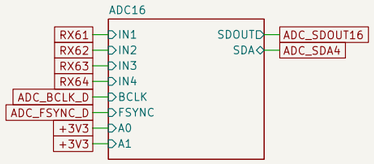












Hybrid 8x8 ultrasonic AESA

Sheet: /ADCs/
File: adcs.kicad_sch

Title: ADCs: 16 x TLV320ADC5140 (4 channels each)

Size: A3	Date: 2026−10−20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 10/14

ADC: TLV320ADC5140 (TI, 4 channels, up to 768 kSPS). Run at 192 kSPS: flat to about 86 kHz.

TDM slave: BCLK 18.432 MHz, FSYNC 192 kHz from the FPGA (shared by all 16 ADCs, so all 64 channels sample at the same instant). 4 slots x 24 bits on SDOUT.

I2C: SCL shared, SDA per group of 4 ADCs; A1/A0 set the address 0x4C–0x4F inside the group. Config writes can use I2C broadcast (P0_R2 I2C_BRDCAST_EN).

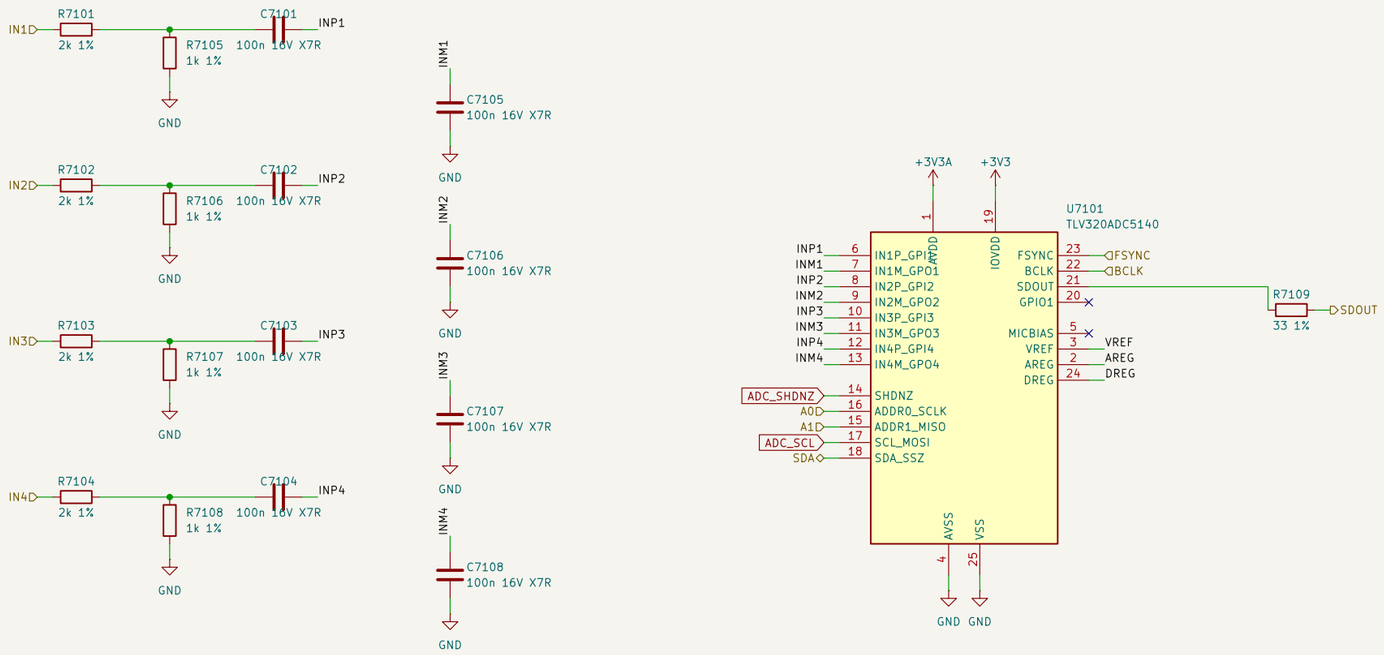
Per channel, set at boot: CHx_INSRC = single-ended (01), CHx_IMP = 10 k (01), AC-coupled, fixed 0 dB gain, DRE and AGC OFF (they change gain on the fly and break channel matching).

Also: ADC_FSCALE 2.75 V (1 Vrms single-ended full scale), AREG_SELECT = 1, TX_EDGE = 1 (the FPGA latches SDOUT one BCLK after launch).

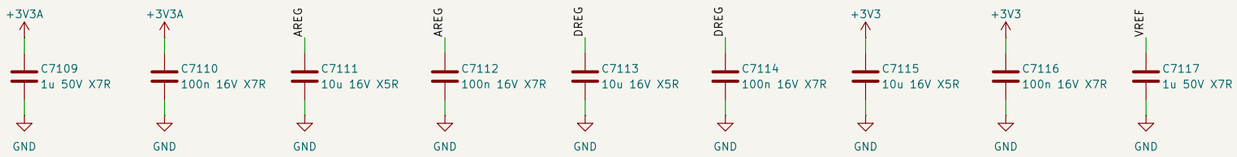
Input: 2k/1k divider (x0.31 with the 10 k input) keeps the receiver's +-4.2 V worst case inside the 1.41 V peak full scale; 100 nF couples to the ADC's own bias (150 Hz high-pass).

Power-up: SHDNZ (ADC_SHDNZ, pulled low on the header sheet) stays low until the rails settle, then wait 2 ms before I2C.

Channel inputs: 2k + 1k divider, 100 nF AC coupling (single-ended, INxM AC-grounded)



Decoupling at the pins (datasheet Figure 172): AVDD 1 uF + 100 nF, AREG and DREG 10 uF + 100 nF, IOVDD 10 uF + 100 nF, VREF 1 uF



Hybrid 8x8 ultrasonic AESA

Sheet: /ADCs/ADC01/

File: adc.kicad_sch

Title: ADC (x16): TLV320ADC5140, 4 channels, 192 kSPS TDM

Size: A3	Date: 2026-10-20	Rev: 2
KiCad E.D.A. 10.0.6		Id: 11/14

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

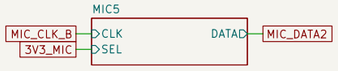
MEMS mics. Mics 2k−1 and 2k share MIC_DATAk−1 (SEL high on the odd one). MIC_CLK_A feeds mics 1–4, MIC_CLK_B mics 5–8.

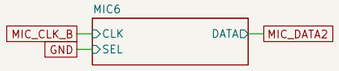


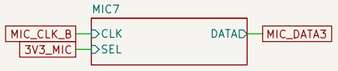


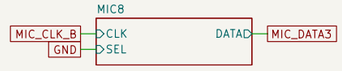












Hybrid 8x8 ultrasonic AESA

Sheet: /MEMS mics/
File: mics.kicad_sch

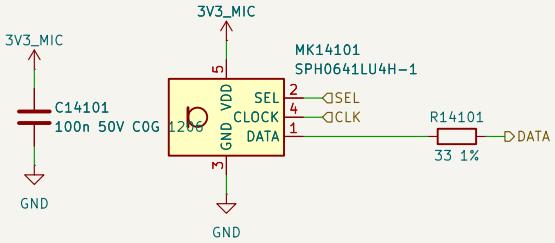
Title: MEMS microphones: 8 x SPH0641LU4H−1

Size: A3	Date: 2026−10−20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 12/14

<< up to: MEMS microphones: 8 x SPH0641LU4H-1 This sheet is used 8 times; the page shows the first instance.

MEMS mic SPH0641LU4H-1 (PDM). Two mics share DATA: SEL high on one, low on the other.

Mounted on the back, sound port through a board hole. 3V3_MIC is the ferrite-filtered rail (power sheet).



Hybrid 8x8 ultrasonic AESA

Sheet: /MEMS mics/MIC1/

File: mic.kicad_sch

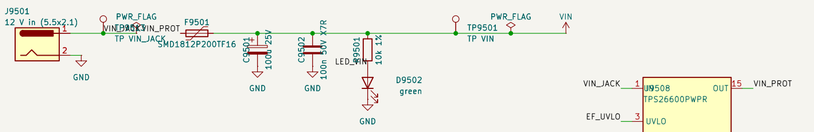
Title: MEMS mic (x8): SPH0641LU4H-1

Size: A4	Date: 2026-10-20	Rev: 2
KiCad E.D.A. 10.0.6		Id: 13/14

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

A. 12 V INPUT: jack -> TPS26600 eFuse (resettable: auto-retry) -> polyfuse (back-up) -> VIN

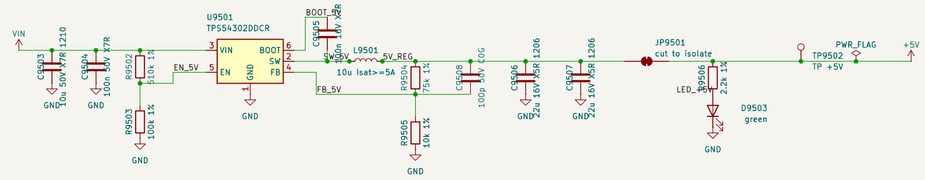
eFuse: input rated 60 V and -60 V (reverse polarity), switches off above 13.8 V (protects the 14.5 V LM2731) and below 8.8 V, current limit 1.6 A.



UVLO/OVP divider 100k / 5.6k / 10k: on 8.8 V, off 13.8 V. ILIM 12 k/A -> 7.5 k = 1.6 A. dVdT 22 nF: 5 V/ms. IMON 72 uA/A into 15 k = 1.08 V/A (TP14). FLT low = fault (TP15).

B. +5V: TPS54302 buck (TI reference design), 3 A

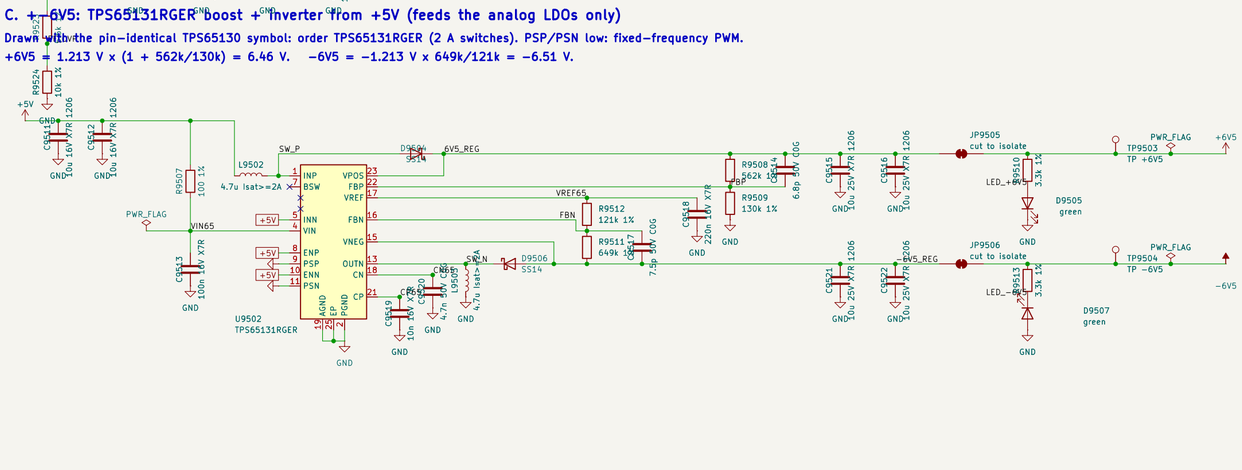
Vout = 0.596 V x (1 + 75k/10k) = 5.07 V. EN UVLO about 7.3 V. Load about 1.6 A.



C. +6V5: TPS65131RGER boost + inverter from +5V (feeds the analog LDOs only)

Drawn with the pin-identical TPS65130 symbol: order TPS65131RGER (2 A switches). PSP/PSN low: fixed-frequency PWM.

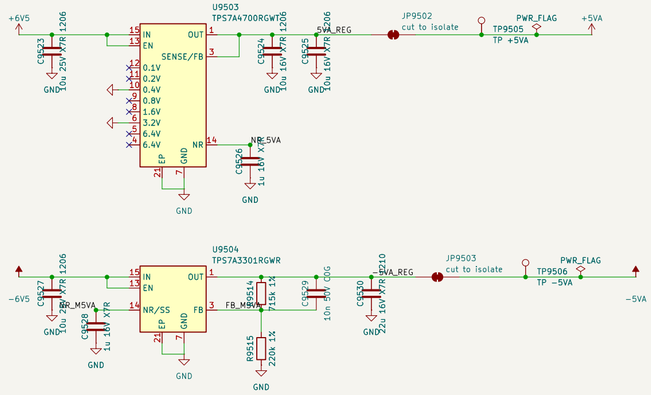
+6V5 = 1.213 V x (1 + 562k/130k) = 6.46 V. -6V5 = -1.213 V x 649k/121k = -6.51 V.



D. ANALOG RAILS: TPS7A4700 +5VA, TPS7A3301 -5VA (low noise, no LEDs)

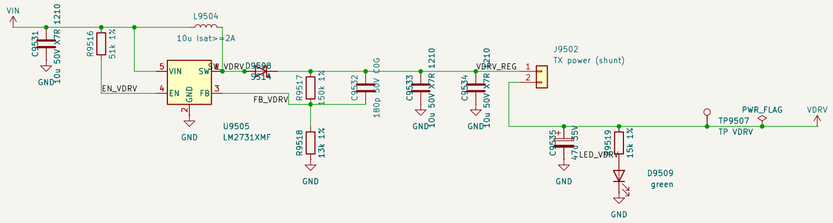
+5VA: ANY-OUT 1.4 + 3.2 + 0.4 = 5.0 V. -5VA = -1.176 V x (1 + 715k/220k) = -5.00 V.

Load: 64 x OPA1678 (4 mA per channel): about 0.3 A on +5VA and on -5VA.



E. VDRV 15 V: LM2731X boost from VIN. Remove the J2 shunt to unpower all TX drivers

1.23 V x (1 + 150k/13k) = 15.4 V. Load, all 64 elements in CW at 40 kHz: about 0.24 A (sim/reports/tx_cw.md).



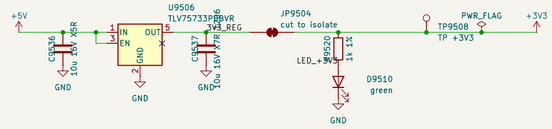
G. GROUND: one GND net and plane

Test points for scope grounds.

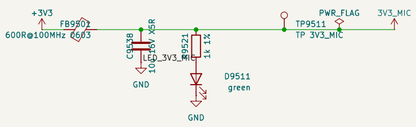


F. +3V3: TLV75733P LDO from +5V (1 A, stable with ceramic caps)

Load < 0.15 A: ADC IOVDD, clock buffers, logic, mics.

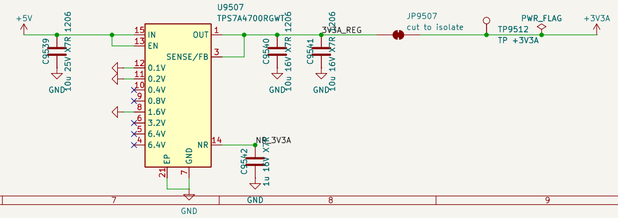


H. 3V3_MIC: ferrite-filtered supply for the 8 MEMS mics (remove FB1 to isolate)



I. +3V3A: TPS7A4700 low-noise LDO from +5V for the 16 ADCs' AVDD (analog rail: no LED)

ANY-OUT 1.4 + 1.6 + 0.2 + 0.1 = 3.3 V. Load about 0.45 A (16 x TLV320ADC5140); 0.8 W in the LDO, thermal vias under the pad.



Hybrid 8x8 ultrasonic AESA

Sheet: /Power/

File: power.kicad_sch

Title: Power: 12 V in, +5V, +-6V5, +-5VA, VDRV, +3V3, 3V3_MIC, +3V3A

Size: A2	Date: 2026-10-20	Rev: 2
KiCad E.D.A. 10.0.6		Id: 14/14